

sample. In the following Sections 5.3.1 and 5.3.2, the concept of tolerance interval, abridged as TI, is used the same way as it is classically presented in the statistical literature. For the validation of analytical methods, two kinds of TI have been proposed:

β -Expectation tolerance interval (β -ETI)	It contains the value of a future observation, with a coverage probability β . As already mentioned, this is the type of statistical interval chosen by the SFSTP commission for the calculation of a MAP and the validation of analytical methods (see Tables 5.3 and 5.4). It corresponds to the letter A in Table 5.5. As explained in Section 7.2, its great advantage is to be suitable for estimating MU since it corresponds to an interval containing a given proportion of the possible measurements of a measurand. This property fits the definition of the coverage interval introduced in Section 7.6.
β - γ Content tolerance interval (β - γ -CTI)	It is also known as the guaranteed coverage tolerance interval with a confidence level of γ [27]. In this case, β is the proportion as in β -ETI, while γ is a confidence level. As specified in ISO 3534-1:2006, in this context, the confidence level is the “long-term proportion of intervals” constructed in this manner that will include at least the expected proportion. This type of statistical dispersion interval is more difficult to relate to the estimation of MU. On the other hand, it is adequately adapted to setting up a quality control or a control chart, as illustrated in Section 7.3.

Tolerance interval estimators generally assume specific data distributions, such as Normal distribution exemplified by Figure 5.9, or Poisson distribution. The data collected to build a MAP has a more complex hierarchical structure due to the fact that measurements are completed in several series. The series is interpreted as a random effect factor, similar to the laboratory effect during an interlaboratory study (see Section 3.2). Therefore, the calculation of the TIs must take this structure into account. Each measurement value is considered as the combination of several distribution laws. The method for computing β -ETI described in the following chapters applies to this type of structured data and has been specifically developed by several authors for balanced and unbalanced experimental designs [27, 28].

5.3.1 β -Expectation Tolerance Interval (β -ETI)

Following our previous convention, X represents the known concentration of a validation material, Y the instrumental response, and Z the inverse-predicted concentration, directly obtained or by inverting the calibration function. The β -ETI for a population of measurement values Z is then expressed as follows:

$$[\bar{\bar{Z}} \pm k_{TI} \times s_{TI}]$$

with

$\bar{\bar{Z}}$: Grand mean or global average of all data ($I \times J$).

k_{TI} : Coverage factor of β -ETI depending on the coverage probability $\beta\%$.

s_{TI} : (Combined) standard deviation of the β -ETI.

The following formulas are applicable for calculating the parameters of β -ETI:
Grand mean of inverse-predicted concentrations

$$\bar{\bar{Z}} = \frac{\sum_{i=1}^I \sum_{j=1}^J Z_{ij}}{I \times J} \quad (5.2)$$

Number of series

$$1 \leq i \leq I \quad (5.3)$$

Number of replicates per series

$$1 \leq j \leq J \quad (5.4)$$

Variance ratio

$$A = \frac{s_B^2}{s_r^2} \quad (5.5)$$

Weighting coefficient Q

$$Q = \frac{A + 1}{J \times A + 1}$$

$$Q = IJ \left(\frac{\frac{s_B^2}{s_r^2} + 1}{J \times \frac{s_B^2}{s_r^2} + 1} \right) \quad (5.6)$$

Variance of intermediate precision (see note 1)

$$s_{IP}^2 = s_r^2 + s_B^2$$

Variance of the β -ETI

$$s_{TI}^2 = s_{IP}^2 \left(1 + \frac{1}{I \times J \times Q} \right)^2 \quad (5.7)$$

Standard deviation of the β -ETI (see note 2)

$$S_{TI} = s_{IP} \sqrt{1 + \frac{1}{I \times J \times Q}} \quad (5.8)$$

Number of degrees of freedom (df) or Number of effective measures

$$N_E = \frac{(A + 1)^2}{\left(\frac{A + \frac{1}{J}}{I - 1} \right)^2 + \frac{1 - \frac{1}{J}}{IJ}} \quad (5.9)$$

$$\text{Coverage factor (quantile of Student's law)} \quad k_{TI} = t_{N_E, \frac{1+\beta}{2}} \quad (5.10)$$

Note 1: Formulas of variances s_{IP}^2 , s_r^2 and s_B^2 are available in Section 3.4.1 and obtained with the classic standard ANOVA algorithm of ISO 5725.

Note 2: The standard deviation of the β -ETI is introduced here for convenience to simplify further computation but does not appear in the literature.

The parameter N_E is considered as a number of *effective* measures because it can be used to check whether the total number of data $I \times J$ collected to build the interval is appropriate, as explained in Section 5.4.3. This number depends on many

intermediate parameters, mainly the variance ratio A . On the other hand, from a statistical viewpoint, it corresponds to the number of degrees of freedom after estimating the TI standard deviation. By definition, it can be interpreted as the number of independent values that remain after all the relationships that link data have been established. It plays a key role in computing the coverage factor k_{TI} .

Because the starting point is to compute the repeatability and intermediate precision variances, Resource E worksheet should be used to compute the β -ETI. The simplest solution is to add new lines and new formulas, leading to another resulting worksheet called Resource H. It is applied to the THEOPHYLLINE dataset on the subset of measurements obtained for the second validation material and listed in Table 5.2. In this example, the assigned reference value is identified as the known concentration of the validation material, that is 0.100 $\mu\text{g/l}$. For each validation material (or level), the same worksheet must be copied and updated with corresponding measurements.

The value in cell B2 (yellow highlighted) containing the assigned value must be consequently modified, as well as the measurement values in cells B5:D10. Following the principle already explained, formulas are made visible in column C to allow the reader to adapt their own worksheet. They are on the right of column B, which contains the expected result. This worksheet is an example that can be improved or adapted. For instance, if there are more than six series of two replicates or if the design is unbalanced. Until row 30, all formulas are the same as in Resource E worksheet. The lines after row 42 will be explained in Section 7.2.4 about MU.

The clumsiest part of this worksheet is between rows 35 and 38, where the coverage factor is computed. Because the formula for obtaining the number of degrees of freedom (df) N_E is complex, it is recommended to split it into several lines to better control possible typing errors.

After all, this number of df is usually non-integer; for this example, it is 9.59. Equation (5.10) shows how the df is used to calculate the quantile of the Student's t distribution law which corresponds to the interval coverage factor. Unfortunately, the Excel built-in function `TINV` which gives the quantile of the Student's t , only accepts integer numbers of df . However, it is essential to know this quantile to obtain the coverage factor.

In the absence of a suitable function, an approximate value should be obtained by linear interpolation between the quantiles given by the upper and lower-rounded integers of N_E . In Section 5.4.3, Figure 5.15 illustrates the differences between exact and approximate Student's t value obtained by linear interpolation. Obviously, this approximation may generate an error that is significant if $N_E < 4$.

We will see that the minimal requirement for building a MAP is to collect at least $I \times J = 9$ measures by level, and it is exceedingly rare to cope with this downside. In such situations, it is recommended to use the `t.ppf` Python function available after importing the `scipy.stats` package that gives exact values for non-integer numbers of df . Lines 42–45 include formulas to calculate the MU. Details about this new quantity are available in Section 7.2, where all equations are explained and commented.

Resource H β -Expectation Tolerance Interval (Excel).

	A	B	C	D	E	F
1	Resource H: β -Expectation Tolerance Interval					
2	Assigned Reference Value ($\mu\text{g/L}$)	0.100				
3		Recovered concentration				
4		Replicate 1	Replicate 2	n(i)	SS(i)	
5	Series 1	0.114	0.113	2	5.00E-07	
6	Series 2	0.112	0.110	2	2.00E-06	
7	Series 3	0.104	0.101	2	4.50E-06	
8	Series 4	0.100	0.105	2	1.25E-05	
9	Series 5	0.113	0.107	2	1.80E-05	
10	Series 6	0.147	0.112	2	6.13E-04	
11	General parameters					
12	Number of series (I)	6	=COUNT(B5:B10)			
13	Number of measures (IJ)	12	=COUNT(B5:C10)			
14	Number of replicates (J)	2	=IF(D5*B12<>B13;"Error";D5)			
15	Residual sum of squares (SSW)	0.00065000	=SUM(E5:E10)			
16	Total sum of squares (SSt)	0.00163500	=DEVSQ(B5:C10)			
17	Inter-series sum of squares (SSB)	0.00098500	=B16-B15			
18	Repeatability variance (s^2_r)	0.00010833	=B15/(B13-B12)			
19	Temporary between-variance	0.00004433	=((B17/(B12-1))-B18)/B14			
20	Between-series variance (s^2_B)	0.00004433	=IF(B19<0;0;B19)			
21	Reproducibility variance (s^2_R)	0.00015267	=B18+B20			
22	Precision					
23	Recovered concentration	0.112	=AVERAGE(B5:C10)			
24	Repeatability std. dev. (sr)	0.0104083	=SQRT(B18)			
25	Between-series std. dev. (sB)	0.0066583	=SQRT(B20)			
26	Intermediate Precision std. dev.. (sFI)	0.0123558	=SQRT(B21)			
27	Trueness					
28	Relative bias (%)	11.5%	=(B23/B2)-1			
29	Recovery yield (%)	111.5%	=B23/B2			
30	β -Expectation Tolerance Interval					
31	Tolerance (beta)	80%				
32	Variance Ratio (A)	0.409	=B20/B18			
33	Coefficient Q	0.7750	=(B32+1)/(B14*B32+1)			
34	Weighing factor W	1.0524	=SQRT(1+1/(B13*B33))			
35	Number of degrees of freedom (NE)	9.59	=(B32+1)^2/(((B32+1/B14)^2/(B12-1)+(1-1/B14)/B13)			
36	t Student lower	1.38	=TINV(1-B31;ROUNDDOWN(B35;0))			
37	t Student upper	1.37	=TINV(1-B31;ROUNDUP(B35;0))			
38	Coverage factor (kTI)	1.38	=B36-(B36-B37)*(B35-ROUNDDOWN(B35;0))			
39	b-ETI standard deviation (sTI)	0.013003	=B26*B34			
40	Lower bond expectation interval	0.094	=B23-B38*B39			
41	Upper bond expectation interval	0.129	=B23+B38*B39			
42	Uncertainty					
43	Composed standard uncertainty (uc(Z))	0.013	=B39			
44	Extended uncertainty (U(Z))	0.026	=2*B43			
45	Relative uncertainty (UR%)	26.0%	=B44/B2			

5.3.2 β - γ Content Tolerance Interval (β - γ -CTI)

Another type of statistical dispersion interval is called β - γ -CTI and is considered by several authors to be well-suited to validate analytical methods [27]. The benefit of β - γ -CTI is to allow an easy computation of the warning and control limits of a control chart, as explained in Section 4.4, but the downside is that it is not suitable for estimating MU. The formulas for the β - γ -CTI are combined below, using the same notations as β -ETI.

Table 5.6 THEOPHYLLINE – parameters of β - γ -CTI ($\beta\% = 80\%$, $\gamma\% = 95\%$).

Reference value ($\mu\text{g/l}$)		0.05	0.10	0.50	1.00	2.50	10.00
Std. dev. of the interval	s_{IC}	0.0214	0.0227	0.0642	0.1634	0.4555	0.8940
Weighting coefficient	W	7.22	9.30	7.23	6.41	12.93	8.91
Coverage factor	k_{IC}	1.37	1.35	1.37	1.38	1.33	1.35
Content interval bounds		0.03	0.08	0.43	0.78	1.91	9.14
		0.09	0.14	0.61	1.23	3.12	11.56

$$\text{General formula } \bar{Z} \pm k_{IC} \times s_{IC} \quad (5.11)$$

$$\text{Coverage factor of the interval } k_{IC} = z_{\left(\frac{1+\beta}{2}\right)} \times \sqrt{1 + \frac{1}{W}} \quad (5.12)$$

$$\text{Quantile of Normal law } z_{\left(\frac{1+\beta}{2}\right)} \quad (5.13)$$

Standard deviation of the interval

$$s_{IC} = \sqrt{s_{FI}^2 + \sqrt{\left(\frac{H_1 \times s_B}{J}\right)^2 + \left(\frac{H_2 \times (J-1) \times s_r}{J}\right)^2}} \quad (5.14)$$

$$\text{Weighting coefficient } W = \frac{I \times \left\{ \left(\frac{SCE_B}{II-1} \right) + (J-1) \left(\frac{SCE_r}{I-1} \right) \right\}}{\left(\frac{SCE_B}{II-1} \right)} \quad (5.15)$$

$$\text{Coefficient } H_1 \quad H_1 = \frac{I-1}{\chi_{1-\gamma, I-1}^2} - 1 \quad (5.16)$$

$$\text{Coefficient } H_2 \quad H_2 = \frac{I(J-1)}{\chi_{1-\gamma, I(J-1)}^2} - 1 \quad (5.17)$$

The notation $\chi_{1-\gamma, I(J-1)}^2$ represents the quantile of the χ^2 distribution (chi-square) for the probability $1 - \gamma$ and the number of degrees of freedom equal to $I(J - 1)$. This value is directly obtained with the Excel built-in function **CHI INV**. Table 5.6 groups the parameters obtained from the THEOPHYLLINE data provided in Table 5.2. Likewise, the β -ETI, before plotting these new intervals on the accuracy profile, the values relative to the reference concentration value X must be calculated. In Table 5.7 table, the relative limits of the β - γ -CTI as well as those of β -ETI and the acceptance limits, are collected for comparison.

The complete MAP for theophylline is illustrated in Figure 5.10. The bounds of the β - γ -CTI are larger as they bring in a confidence level. They can be interpreted as intervals where the *true* β -ETI may probably lie with a risk of error of 5%. The worksheet about precision parameters referenced Resource E was extended to give a new worksheet applicable to the computation of the β -ETI and called Resource H. Likewise, the worksheet called Resource I β - γ Content Tolerance Interval (Excel) for the β - γ -CTI is an extension of Resource H and uses the same layout.